Probability from Observation:
Carathéodory and Stone Routes to the Observable Extension Theorem

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Abstract

We prove that a coherent family of observations determines a unique probability measure on the observable σ-algebra. The setting is a directed system of measurable outcome spaces connected by surjective refinement maps, together with a compatible family of finitely additive charges on the resulting cylinder algebra. We identify the necessary and sufficient condition for the family to extend to a σ-additive global measure: *collective exhaustion* (CE), the condition that no level can assign persistent mass to events the system as a whole sees as empty.

The argument proceeds by two independent routes, each illuminating a different aspect of the extension problem. The *Carathéodory route* assumes σ-additive marginals and constructs the global measure by a direct Carathéodory argument on the realization space, with no topological hypotheses on the outcome spaces. The *Stone route* assumes only finite additivity and derives σ-additivity from the compactness of the Stone space: the directed system of Boolean algebras dualises to an inverse system of compact Hausdorff spaces, whose inverse limit carries a canonical σ-additive measure; CE then ensures this measure concentrates on the image of the sample space inside its Stone compactification. Under shared hypotheses the two measures coincide.

A foundational analysis shows that CE is *irreducible*: it is not derivable from any finitary or structural condition on the query system. The finite-cofinite counterexample and a model-theoretic argument via Łoś's theorem locate the boundary precisely. CE is not an assumption imposed from outside but the exact condition separating directed systems in which every convergence is witnessed somewhere in the refinement order from those in which it is not.

Beyond the extension theorem, the Stone space $\mathrm{St}(\mathcal{C})$ constructed here as a technical device has further significance: it is the canonical compact Hausdorff completion of the sample space for the observable topology. When an observation function is a cyclic vector for the Koopman operator of an underlying dynamical system, this Stone space is measure-theoretically isomorphic to the state space. The compact space introduced here reappears at the end of the programme as the object being reconstructed.

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1 Introduction

Many approaches to probability begin by postulating an underlying probability space and then studying the observables derived from it. This paper reverses that perspective. Starting from observable experiments organised as a refinement system of queries, we ask: under what conditions do compatible observable laws determine a canonical probability measure?

The answer is given by a single condition. A compatible family of finitely additive charges on a directed system of Boolean algebras extends to a σ-additive measure at every level if and only if the family is *collectively exhaustive* (CE): for every decreasing sequence of events whose intersection is empty, some level in the refinement order witnesses the vanishing of mass. CE is not a structural condition on the query system but a condition on the charges: it rules out purely finitely additive charges, which satisfy every finite consistency condition while assigning persistent mass to sequences that vanish in the limit.

Two routes. The extension theorem is established by two independent proofs, each illuminating CE from a different direction.

The *Carathéodory route* (§4) assumes σ-additive marginals and constructs the global measure directly by Carathéodory extension on the realization space. No topology is required on the outcome spaces. Sequential common refinements compress any countable cylinder family to a single query level, reducing σ-subadditivity of the cylinder premeasure to σ-subadditivity of a single marginal. In this route, σ-additivity of the marginals is the standing hypothesis; by the CE theorem of §3, it is equivalent to CE.

The *Stone route* (§5) assumes only finite additivity and derives σ-additivity from compactness. Stone duality converts the directed system of Boolean algebras into a cofiltered inverse system of compact Hausdorff spaces. A compatible family of finitely additive charges assembles into a σ-additive measure on the inverse limit without any countable additivity assumption. The remaining task is to descend from the Stone space to the sample space. Two conditions are required. Discriminability (§2) ensures the Stone embedding is injective, so the pullback identifies the observable σ-algebra with the Borel algebra of the Stone space. CE appears here not as an algebraic condition but as a support condition: the measure concentrates on the principal ultrafilters, i.e., on the image of Ω inside its Stone compactification.

CE is irreducible. Section 3 establishes that CE is not derivable from any finitary or structural condition on the query system. The finite-cofinite content on $\mathbb{Q}$ satisfies every algebraic condition on a directed system while failing CE. A model-theoretic argument via Łoś's theorem and the Yosida–Hewitt decomposition shows this is not a proof-search failure: no first-order condition on the query system implies CE. CE is an irreducible commitment about the completed infinite, analogous to the axiom of infinity in arithmetic.

The Stone space. The Stone space $\mathrm{St}(\mathcal{C})$ of the observable algebra, constructed in §5 as a technical device for measure extension, has a further significance developed in §6. It is the canonical compact Hausdorff completion of Ω for the observable topology. CE's two faces — honest observer (valuation layer) and support on Ω (topological layer) — are the same condition seen from inside and outside the algebra. When an observation function is a cyclic vector for the Koopman operator of an underlying dynamical system, the Stone space of the observable algebra is measure-theoretically isomorphic to the state space. The compact space introduced here as a tool reappears at the end of the programme as the object being reconstructed; we develop this connection elsewhere.

Organisation. Section 2 introduces query systems, cylinder sets, the observable σ-algebra, compatible charges, and the three hypotheses (EvalSurjective, Discriminability, CE). Section 3 establishes the CE theorem and its irreducibility. Section 4 proves the Carathéodory route. Section 5 proves the Stone route. Section 6 draws the two routes together and discusses the Stone space.

2 Setup: query systems and charges

2.1 Query systems

Definition 2.1 (Query system)**.** A *query system* is a tuple

$$(\iota, \leq, \Omega, \{(\mathrm{Out}(i), \mathcal{F}_i)\}_{i\in\iota}, \{\mathrm{eval}_i\}_{i\in\iota}, \{\pi_{ij}\}_{i\leq j})$$

consisting of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- a set Ω, the *sample space*;
- for each $i \in \iota$, a measurable space $(\mathrm{Out}(i), \mathcal{F}_i)$, the *outcome space at level* i;
- for each $i \in \iota$, a map $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$, the *evaluation map at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathrm{Out}(j) \to \mathrm{Out}(i)$, the *refinement map*,

satisfying the *coherence condition*: $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$.

The index i represents a level of observational resolution. When $i \leq j$, level j is finer: its outcomes can be coarsened to level i by applying π_{ij}. The coherence condition says that the evaluation maps are compatible with the coarsening structure.

2.2 Realization space

When the query system is constructed from a projective system of outcome spaces, the sample space Ω is the realization space

$$\Omega = \varprojlim_{i \in \iota} \mathrm{Out}(i) = \Big\{ x \in \prod_{i \in \iota} \mathrm{Out}(i) \;\Big|\; \pi_{ij}(x_j) = x_i \text{ whenever } i \leq j \Big\},$$

with $\mathrm{eval}_i(x) = x_i$.

2.3 Directedness conditions

Definition 2.2 (Common coarsenings and common refinements)**.** The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$. It admits *common refinements* if every pair has an upper bound: there exists $k \geq i$ and $k \geq j$. It admits *sequential common refinements* if every countable family $\{i_n\}_{n \geq 1} \subset \iota$ has an upper bound in ι.

Sequential common refinements imply common refinements. Common coarsenings ensure that cylinder sets form a π-system (Lemma 2.4 below); sequential common refinements supply the σ-subadditivity step in the Carathéodory route.

2.4 Cylinder sets and the observable σ-algebra

Definition 2.3 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{F}_i$, the *level-*i *cylinder with base* A is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder generating family* is

$$\mathcal{C} \;=\; \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{F}_i\}.$$

For each $i \in \iota$, let B_i denote the Boolean algebra of subsets of Ω generated by $\{\mathrm{Cyl}(i, A) : A \in \mathcal{F}_i\}$. Then $\mathcal{C} = \bigcup_{i \in \iota} B_i$.

The *observable σ-algebra* is $\sigma(\mathcal{C})$, the σ-algebra on Ω generated by all cylinder sets.

The coherence condition gives, for $i \leq j$, $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$, so the same subset of Ω can be described as a cylinder at any finer level.

Lemma 2.4 (Cylinder π-system)**.** *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound, with $k \leq i$ and $k \leq j$. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

2.5 Connecting maps

For $i \le j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.6 Compatible charges

Definition 2.5 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \le j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$; compatibility ensures this is well-defined.

2.7 The three hypotheses

The main theorems require three conditions.

Definition 2.6 (EvalSurjective)**.** The query system satisfies *EvalSurjective* if $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$ is surjective for every $i \in \iota$.

Definition 2.7 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Definition 2.8 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

EvalSurjective is a structural condition ensuring that every outcome at every level is realized by some element of Ω; it implies that the refinement maps π_{ij} are surjective (by coherence) and that the connecting maps φ_{ij} are injective. Discriminability ensures the Stone embedding $\omega \mapsto \{\text{principal ultrafilter at } \omega\}$ is injective. CE is the irreducible valuation-layer condition studied in §3; it rules out purely finitely additive charges and is the necessary and sufficient condition for σ-additive extensibility.

3 The CE theorem

This section establishes collective exhaustion as the necessary and sufficient condition for σ-additive extensibility. The analysis works at the level of an abstract directed system of Boolean algebras; it does not require the full query-system structure of §2. To keep this generality clean, we write $\mathcal{E}_i$ for a generic Boolean algebra at level i and ℓ_i for the charge on it. The connection to query systems is: $\mathcal{E}_i = B_i$ (the Boolean algebra of cylinder sets at level i), $O_i = \mathrm{Out}(i)$, and $\ell_i = \mu_i$. The full query system structure re-enters in §4 and §5.

3.1 Single-algebra extension

Definition 3.1 (Boolean charge space)**.** A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation. No σ-algebra and no topology are assumed on O.

Under Stone's representation theorem [11], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The *gap* $\sigma(\mathcal{E}) \setminus \mathcal{E}$ consists of the Baire sets that are not clopen: events that countable operations can reach but that no single observable distinction can name.

Theorem 3.2 (Extension criterion)**.** *A normalized finitely-additive $\ell : \mathcal{E} \to [0,1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$. The extension is then unique.*

Proof. Standard; see Halmos [6] or Fremlin [5] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ-additivity on $\sigma(\mathcal{E})$. □

Remark 3.3 (The gap and the regress)**.** The condition in Theorem 3.2 is not verifiable from within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope that a finer level $j \geq i$ can witness the emptiness that level i cannot see, and that compatibility forces continuity at $\emptyset$ at level i. But the obstruction re-appears: finite additivity at level j does not force continuity at $\emptyset$ for sequences decreasing to an element of $\sigma(\mathcal{E}_j)$. No finite chain of refinements can supply the condition. Something external to any single level is required: topology (compactness), a colimit algebra, or an explicit continuity axiom. Section 3.2 shows that the directed-system structure does not resolve this without an additional valuation condition.

3.2 Nontrivial refinement

Definition 3.4 (Nontrivial refinement)**.** Let $\pi : O_j \to O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 3.5 (Nontrivial refinement introduces new events)**.** *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. □

Proposition 3.6 (Gap nonemptiness)**.** *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ-algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ-algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in the gap. □

3.3 CE characterises σ-additivity

We now state the directed-system version of CE and prove the main equivalence.

Definition 3.7 (Collectively exhaustive family)**.** A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$.

Here $\pi_{ij} : O_j \to O_i$ is the refinement map (j finer than i), so $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of $E_n \subseteq O_i$ at the finer level.
CE is a *valuation-layer* condition: it places a requirement on how the charges behave, not on the Boolean-algebraic index structure. It says that no level can assign persistent mass to events that the system as a whole sees as empty.

The following proposition establishes that the index-layer structure of a directed system — sequential upper-directedness and charge compatibility — does not on its own force σ-additivity.

Proposition 3.8 (Index-layer conditions do not force σ-additivity)**.** *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. Let $\iota = \mathbb{N} \cup \{\omega\}$ with $n \leq \omega$ for all $n \in \mathbb{N}$, directed by this order. Take all outcome spaces to be $\mathbb{Q}$, all event algebras to be the finite-cofinite algebra $\mathcal{E} = \{E \subseteq \mathbb{Q} : E \text{ is finite or } E^c \text{ is finite}\}$, all refinement maps to be the identity, and $\ell(E) = 0$ if E is finite, $\ell(E) = 1$ if E is cofinite. This is a normalized finitely-additive content and compatibility holds trivially.

Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$ and let $F_n = \{q(0), \dots, q(n)\}$. Each F_n is finite so $\ell(F_n) = 0$, yet $\bigcup_n F_n = \mathbb{Q}$ and $\ell(\mathbb{Q}) = 1$. σ-subadditivity would give $1 \leq \sum_n \ell(F_n) = 0$: contradiction. □

Theorem 3.9 (CE characterises σ-additivity)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) *The family is collectively exhaustive.*

(ii) *ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.*

Proof. (i) ⇒ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. Theorem 3.2 gives the unique σ-additive extension.

(ii) ⇒ (i). Suppose ℓ_i extends to μ_i for every i. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization. Take $j = i$. □

Remark 3.10 (Dissolution of the index-valuation conflation)**.** Proposition 3.8 shows that index-layer conditions (sequential upper-directedness, compatibility) do not force σ-additivity. Theorem 3.9 identifies the exact valuation-layer condition that does. The question "do the structural conditions on a directed system force σ-additivity?" is ill-posed: it conflates conditions on the directed system's structure with conditions on the charges it carries.

Sequential upper-directedness plays no role in Theorem 3.9: it is a per-level result. That condition enters when one asks whether the per-level extensions assemble into a global measure on the realization space $\Omega = \varprojlim \operatorname{Out}(i)$; this is the role it plays in the Carathéodory route (§4).

3.4 CE is irreducible

Proposition 3.11 (CE irreducibility)**.** *Collective exhaustion is not implied by any structural condition on a query system.*

(i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*

(ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 3.8: the finite-cofinite content satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś's theorem [9]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$ (using the enumeration from Proposition 3.8). Each δ_n is a normalized

σ-additive content satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n)\in E}$, which is 1 on cofinite sets and 0 on finite sets — exactly the finite-cofinite content ℓ from Proposition 3.8.

Step 3. By Step 1, any universally valid first-order Φ holds for ℓ. But ℓ fails CE. Therefore no first-order Φ can imply CE.

The Yosida–Hewitt decomposition [13] explains why this obstruction is permanent. Every finitely-additive probability on a Boolean algebra decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive*: it satisfies every algebraic condition while assigning zero to every singleton. The finite-cofinite content ℓ is purely finitely additive ($\ell_c = 0$). CE is exactly the condition that rules out a nonzero purely finitely additive part: $\ell_p = 0$. □

Remark 3.12 (CE as irreducible primitive)**.** Proposition 3.11 shows that the gap between structural coherence and σ-additivity is a proved boundary, not a proof-search failure. CE is not derivable within the query-system framework. It is analogous to the axiom of infinity in arithmetic: not provable from the remaining axioms, not refutable, but the commitment without which the mathematics of probability cannot proceed. The ultrafilter-based content is the canonical witness of this irreducibility: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty.

Formally: CE is not a constraint the query system imposes on the observer. It is a commitment the observer makes about the completed infinite, a condition on how their charges behave with respect to limits that no finite stage can verify.

4 Route 1: The Carathéodory extension

This section establishes the extension theorem by a direct Carathéodory argument. The input is a query system satisfying EvalSurjective, sequential common refinements, and common coarsenings, together with a compatible family of σ-additive marginals. By Theorem 3.9, the σ-additivity of the marginals is equivalent to CE applied at each level.

4.1 Observational determination

Theorem 4.1 (Observational determination)**.** *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.4 the cylinder family $\mathcal{C}$ is a π-system generating $\sigma(\mathcal{C})$. The π–λ theorem gives $P = P'$. □

4.2 Observable content

Theorem 4.2 (Finite observational content)**.** *Assume the query system admits common coarsenings and common refinements and satisfies EvalSurjective, and let $\{\nu_i\}_{i\in\iota}$ be an observationally compatible family with each $\nu_i \in \mathcal{P}(\mathrm{Out}(i))$. Then the assignment*

$$\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A), \qquad A \in \mathcal{F}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in $\mathrm{Out}(k)$ are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and EvalSurjective (surjectivity of eval_k) imply these coincide in $\mathrm{Out}(k)$. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k. □

4.3 Carathéodory extension theorem

Theorem 4.3 (Observational extension theorem)**.** *Suppose:*

(i) $(\iota, \leq)$ *admits common coarsenings and sequential common refinements;*

(ii) the query system satisfies EvalSurjective;

(iii) the family $\{\nu_i\}_{i\in\iota}$ *is observationally compatible and each* ν_i *is a probability measure on* $(\mathrm{Out}(i), \mathcal{F}_i)$*.*

Then there exists a unique probability measure P *on* $(\Omega, \sigma(\mathcal{C}))$ *such that* $(\mathrm{eval}_i)_\# P = \nu_i$ *for every* $i \in \iota$*.*

Proof. Step 1: Finite content. Theorem 4.2 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \ldots\}$ to find a single index k with $k \geq i$ and $k \geq i_n$ for all n. Compress each cylinder to level k:

$$\mathrm{Cyl}(i, B) = \mathrm{Cyl}(k, E), \qquad \mathrm{Cyl}(i_n, A_n) = \mathrm{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. EvalSurjective and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in $\mathrm{Out}(k)$. Therefore

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a semiring generating $\sigma(\mathcal{C})$ and μ_0 is a σ-subadditive finitely additive content on it. Carathéodory's extension theorem [1, 2] applies and yields P on $(\Omega, \sigma(\mathcal{C}))$.

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{F}_i$, $P(\mathrm{Cyl}(i, A)) = \mu_0(\mathrm{Cyl}(i, A)) = \nu_i(A)$, so $(\mathrm{eval}_i)_\# P = \nu_i$. Setting $A = \mathrm{Out}(i)$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Theorem 4.1. □

Remark 4.4 (Relation to Kolmogorov's extension theorem)**.** Kolmogorov's theorem [8] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 4.3 plays the same role for query systems: the realization space $\Omega = \varprojlim \mathrm{Out}(i)$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

Remark 4.5 (Prokhorov route)**.** A third route to the extension theorem starts from *finitely* additive marginals and derives σ-additivity from topological compactness. Under *surjective projective uniform tightness* — a condition requiring that for every $\varepsilon > 0$ there exist compact sets $K_i \subset \mathrm{Out}(i)$ with $\nu_i(\mathrm{Out}(i) \setminus K_i) < \varepsilon$ and $\pi_{ij}(K_j) = K_i$ for all $i \leq j$ — the Prokhorov theorem extends finitely additive marginals on a countable sequentially upper-directed query system with Hausdorff outcome spaces to a σ-additive global measure. This route assumes topological structure on the outcome spaces and is complementary to the Carathéodory and Stone routes, which require no topology.

Remark 4.6 (Lean formalization)**.** Theorem 4.3 is fully formalized in Lean 4 (`observational_extension` in `QuerySystem.lean`, 0 sorrys). The hypotheses correspond exactly to the four conditions: `LowerDirected`, `SequentiallyUpperDirected`, `EvalSurjective`, and `CompatibleMarginals` with `IsProbabilityMeasure`. The Carathéodory step uses `AddContent.measure` from Mathlib [12].

5 Route 2: The Stone extension

This section gives the second, independent proof of the extension theorem. The input is weaker: charges need only be finitely additive. σ-additivity is derived from the compactness of the Stone space.

The proof proceeds in four steps. Step A identifies the Stone space of the direct limit with the inverse limit of individual Stone spaces. The inverse limit measure step uses Choksi's theorem to produce a σ-additive measure on the inverse limit. Step B1 identifies the pullback of the Borel algebra of the Stone space with the observable σ-algebra. Step B2 shows that CE forces the measure to concentrate on the image of Ω.

5.1 Step A: Stone space of the direct limit

Lemma 5.1 (Injectivity of connecting maps)**.** *If the query system satisfies EvalSurjective, then $\varphi_{ij} : B_i \to B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. EvalSurjective and the coherence condition give $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega)$ for all ω, so $\mathrm{im}(\pi_{ij}) \supseteq \mathrm{im}(\mathrm{eval}_i) = \mathrm{Out}(i)$: the refinement maps are surjective. A surjective map π_{ij} has injective preimage map π_{ij}^{-1} on subsets, so $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ is injective on generators, hence on B_i. $\square$

Proposition 5.2 (Step A)**.** *The Stone space of the direct limit satisfies* $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [7, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. $\square$

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure)**.** *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [3], §6, or [6], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [4, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. □

5.3 Step B1: the structural identification

Let pure : $\Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 5.4 (Step B1: structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [7, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. □

Remark 5.5. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 Step B2: CE and the support condition

Proposition 5.6 (Step B2: support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on* $\mathrm{St}(\mathcal{C})$ *is supported on* $\mathrm{pure}(\Omega)$*, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [13, 10], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [10], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 3.9). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. □

5.5 The Stone extension theorem

Theorem 5.7 (Stone extension theorem)**.** *Let the query system satisfy EvalSurjective, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{F}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S 5.2,\text{ Step 1}} \{\hat{\mu}_i\} \xrightarrow{\S 5.2,\text{ Step 3}} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\text{B1}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.6, $\hat{\mu}$ is supported on pure(Ω). Since pure is injective (Discriminability), the pushforward $P = (\mathrm{pure})_*\hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 5.4, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Theorem 4.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to EvalSurjective, Discriminability, and CE. ☐

Remark 5.8 (CE is not bypassed by compactness)**.** It may appear that the compactness of the Stone space eliminates the need for CE. Compactness does ensure that $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is σ-additive. But $\hat{\mu}$ lives on the Stone space, not on Ω. The descent from $\mathrm{St}(\mathcal{C})$ to Ω requires that no mass escapes to the non-principal ultrafilters, and that is precisely CE (Proposition 5.6). In the Stone picture, CE is not an additional hypothesis but a *support condition*: the measure concentrates on the image of the sample space inside its Stone compactification.

Remark 5.9 (Lean formalization)**.** The Stone route is formalized in Lean 4 in `StoneDualityExtension.lean`. The structural results (Tasks 0′-A and 0′-C: Stone space, cylinder clopens, evaluation maps, bonding maps) are fully proved with no sorry obligations. Two intentional sorrys mark Mathlib gaps: the extension of a clopen-algebra charge to a regular Borel measure on a compact totally-disconnected space (Halmos §§53–54; Fremlin Vol. 1 §311E), and Choksi's projective-limit theorem for compatible measures on cofiltered inverse systems of compact Hausdorff spaces. The coincidence of routes (Task 0′-E: `stone_agrees_with_caratheodory`) is fully proved, as an immediate consequence of `observational_determination`.

6 The bridge

6.1 Two routes, one theorem

Under shared hypotheses — EvalSurjective, Discriminability, and σ-additive compatible marginals — both routes produce the same measure.

Proposition 6.1 (Coincidence of routes)**.** *Suppose the query system satisfies EvalSurjective, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^{C} be the measure produced by Theorem 4.3 and P^{S} the measure produced by Theorem 5.7. Then $P^{\mathrm{C}} = P^{\mathrm{S}}$.*

Proof. Both measures agree on every cylinder set: $P^{\mathrm{C}}(\mathrm{Cyl}(i, A)) = \nu_i(A) = P^{\mathrm{S}}(\mathrm{Cyl}(i, A))$. By observational determination (Theorem 4.1), they are equal on $\sigma(\mathcal{C})$. ☐

The two routes are genuinely independent proofs of the same theorem. The Carathéodory route works directly within measure theory; the Stone route takes a detour through topology, using compactness of the Stone space to derive σ-additivity that the Carathéodory route assumes. That they produce the same measure is not obvious from the constructions alone; it is a consequence of uniqueness (Theorem 4.1), which is doing real work here.

6.2 CE's two faces

CE appears in the two routes in structurally different forms, yet it is the same condition.

In the Carathéodory route (Route 1), CE appears at the level of the charges: the per-level marginals are σ-additive, which by Theorem 3.9 is equivalent to CE applied at each level. Here, CE is an *algebraic* condition: the charges do not assign persistent mass to events that the observer's refinement hierarchy sees as empty. It is a condition on the observer's commitments.

In the Stone route (Route 2), CE appears as a *topological* condition: the Baire measure $\hat{\mu}$ on the Stone space must be supported on pure(Ω), the image of the sample space inside its compact compactification. Mass on non-principal ultrafilters corresponds exactly to the purely finitely additive part μ_p, and CE $\Leftrightarrow \mu_p = 0$ (Proposition 5.6). Here, CE is a *geometric* condition: the observer's measure does not escape to the "phantom" points of the Stone compactification.

The Yosida–Hewitt decomposition is the bridge between the two faces: it identifies the purely finitely additive part of μ with the mass that has escaped to non-principal ultrafilters. CE says both things simultaneously: the algebraic convergence is honest, and the geometric measure is confined to Ω.

6.3 The Stone space as compact completion

The Stone space $\mathrm{St}(\mathcal{C})$ constructed in §5 as a technical device for measure extension is, in fact, the canonical compact Hausdorff completion of the sample space for the observable topology.

More precisely: the Stone embedding pure : $\Omega \to \mathrm{St}(\mathcal{C})$ is a dense embedding when Ω is dense in $\mathrm{St}(\mathcal{C})$ for the Stone topology — i.e., when every nonempty basic clopen $[E]$ meets pure(Ω), which holds whenever $E \neq \emptyset$ as a subset of Ω. The Stone space is uniquely determined by $\mathcal{C}$ up to homeomorphism. It is not a construction imposed on Ω from outside but the compactification already contained in the Boolean algebra structure of the cylinder sets.

CE is precisely the condition that the measure P on Ω, when viewed as a measure on its compact completion $\mathrm{St}(\mathcal{C})$ via pure, does not escape: it remains concentrated on the original (non-compact) sample space. In this sense, an observer satisfying CE "knows they are in Ω": their probability assignments, however they are extended to the compact hull, retain no phantom mass.

Remark 6.2 (Topology from the query system)**.** The Stone space $\mathrm{St}(\mathcal{C})$ carries a canonical topology: the one generated by all cylinder clopens $[E]$. This is the coarsest topology making every observable distinction continuous. A natural question, deferred to later work, is whether the *relevant* topology on Ω — the one appropriate for dynamics and reconstruction — is this Stone topology, or whether it should itself be derived from the query system and its charges. When separation is weighted by the measure P, a third regime appears beyond the usual distinguishable/indistinguishable binary: pairs that are formally separated at every level but whose separating events have P-measure tending to zero. These are points that the observer's valuation treats as near even though the Boolean algebra keeps them apart. Whether this measure-weighted nearness assembles into a topology, and whether collective exhaustion and this separation condition are two faces of a single coherence requirement, is an open question.

Connection to reconstruction. For a dynamical system (X, T, μ) with observation function h, the time-delay cylinder algebras $\{B_S\}$ (indexed by finite subsets S of delay times) form a directed system of exactly the kind treated here. The Stone space of the resulting observable algebra $\sigma(B_\infty)$ is, when h is a *cyclic vector* for the Koopman operator U_T — i.e., the orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ spans a dense subspace of $L^2(X, \mu)$ — measure-theoretically isomorphic to the state space X itself.

The compact space $\mathrm{St}(\mathcal{C})$ introduced here as a tool for measure extension is, under the reconstruction condition, the state space being reconstructed. The programme begins with a query system whose Stone space is a compact completion of Ω, and ends with the recognition that this compact space is the state space of the dynamical system the observer is embedded in. We develop this connection in a separate paper.

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